

# Chamroeun Hongleng

Computer science student and AI-native learner — full-stack products, backend systems, machine-learning tools.

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Seeking a software engineering internship · Phnom Penh or remote

## PROJECTS

**PhsarOS** — operations platform for small shops · main builder Jul 2026 — present  
*Next.js 15 (App Router, RSC, Server Actions) · TypeScript · Prisma · PostgreSQL · Auth.js · Zod · phsaros.vercel.app*

- Live multi-tenant product: signup, point of sale, inventory, customers, expenses, and a server-computed dashboard. Anyone can sign up and complete a real sale.
- Tenant isolation in the query layer, Zod-validated Server Actions, atomic sales, an eight-model Prisma schema, and Vitest with typechecking in CI. Authored 59 of the project's 63 commits.

**chamroeunhongleng.me** — schema-validated content system · sole builder Aug 2026 — present  
*Nuxt 4 · TypeScript · Zod · design tokens · Vitest · GitHub Actions + CodeQL · Vercel · source (public)*

- Content in JSON validated by Zod schemas, with routes and the sitemap derived from it; one shared rule engine fails production builds on any unresolved placeholder.
- 99 unit tests, a twelve-phase verification pipeline, GitHub Actions and CodeQL, and an assistant whose answers are graded by automated evaluations.

**Kaskor ASR** — Khmer speech recognition · sole builder 2026 — present  
*Python · PyTorch · Hugging Face Transformers · Whisper-small · MIT · github.com/chamroeunhongleng/kaskor-asr*

- Eleven-script pipeline from raw audio to trained model over ~86,500 samples, plus a pip-installable CLI and a published model card.
- Identified and fixed a 225-token evaluation cap that truncated model output, correcting the self-reported CER from 17.48% to 3.74% on a speaker-overlapping validation split.

**Chomkar Decision Grid** — deterministic decision engine · sole builder 2026 — present  
*Python (stdlib only) · unit tests · GitHub Actions · MIT · github.com/chamroeunhongleng/chomkar-decision-grid*

- Audit stage re-derives every result through independent code paths; 62 unit tests over invariants and designed failure paths, run in CI.

## EXPERIENCE

**CHNAI LAB** — student product studio · software developer and organization administrator 2026 — present

- Build and maintain PhsarOS, the studio's shop-operations product, and administer its GitHub organization. Products are divided between six members under shared contribution, security, and AI-use standards.
- On Chomkar, the studio's agritech product: provide the Khmer voice-intake model the team builds against — my own Kaskor ASR, served from my Hugging Face account and used in the hackathon demo — plus ~30 farmer interviews and the business analysis behind its pre-harvest order model.

## SKILLS

|                           |                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------|
| <b>WEB</b>                | TypeScript, Next.js (App Router, RSC, Server Actions), Nuxt 4 / Vue, React, Tailwind                       |
| <b>BACKEND</b>            | PostgreSQL, Prisma, Auth.js with JWT, Zod validation, multi-tenant isolation, serverless                   |
| <b>ML</b>                 | Python, PyTorch, Hugging Face Transformers, Whisper fine-tuning, evaluation design                         |
| <b>AI-NATIVE PRACTICE</b> | AI-assisted development under written repository contracts, AI workflow design, automated model evaluation |
| <b>PRACTICE</b>           | Git, GitHub Actions CI, Vitest and pytest, strict typechecking, ESLint, CodeQL, Vercel                     |
| <b>LANGUAGES</b>          | Khmer — native; English — intermediate (B1–B2), studying B.A. English for Work Skills                      |

## EDUCATION

**American University of Phnom Penh / Fort Hays State University** — dual degree: B.S. Oct 2025 — present  
Information Technology Management and B.S. Computer Science

**Institute of Foreign Languages, RUPP** — B.A. English for Work Skills (international business) Jan 2026 — present